



Clinical, paraclinical and outcome features of 166 patients with acute anti-GQ1b antibody syndrome

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Abstract

Background & purpose In this retrospective study, we aimed at defining the clinical, paraclinical and outcome features of acute neurological syndromes associated with anti-GQ1b antibodies.

Results We identified 166 patients with neurological symptoms appearing in less than 1 month and anti-GQ1b antibodies in serum between 2012 and 2022. Half were female (51%), mean age was 50 years (4–90), and the most frequent clinical features were areflexia (80% of patients), distal upper and lower limbs sensory symptoms (78%), ophthalmoplegia (68%), sensory ataxia (67%), limb muscle weakness (45%) and bulbar weakness (45%). Fifty-three patients (32%) presented with complete (21%) and incomplete (11%) Miller Fisher syndrome (MFS), thirty-six (22%) with Guillain–Barre syndrome (GBS), one (0.6%) with Bickerstaff encephalitis (BE), and seventy-three (44%) with mixed MFS, GBS & BE clinical features. Nerve conduction studies were normal in 46% of cases, showed demyelination in 28%, and axonal loss in 23%. Anti-GT1a antibodies were found in 56% of cases, increased cerebrospinal fluid protein content in 24%, and *Campylobacter jejuni* infection in 7%. Most patients (83%) were treated with intravenous immunoglobulins, and neurological recovery was complete in 69% of cases at 1 year follow-up. One patient died, and 15% of patients relapsed. Age > 70 years, initial Intensive Care Unit (ICU) admission, and absent anti-GQ1b IgG antibodies were predictors of incomplete recovery at 12 months. No predictors of relapse were identified.

Conclusion This study from Western Europe shows acute anti-GQ1b antibody syndrome presents with a large clinical phenotype, a good outcome in 2/3 of cases, and frequent relapses.

Keywords Anti-GQ1b antibodies · Miller-Fisher syndrome · Bickerstaff’s brainstem encephalitis · Guillain–Barre syndrome

Introduction

Gangliosides, including GQ1b, are glycosphingolipids with a structure in which a hydrophobic ceramide and a hydrophilic sugar chain portion are bound, and sialic acid is bound to the sugar chain portion [1, 2]. Gangliosides are abundant in the nervous system, and their distribution is heterogeneous between different nerve tissues [2]. GQ1b represents approximately 5–6% of the total gangliosides in human peripheral nervous system (PNS), and 11–13%

in oculomotor nerves [2]. In oculomotor nerves, GQ1b are mostly localized in the paranodal region of the node of Ranvier [2]. In addition to oculomotor nerves, the GQ1b sugar chain antigen is also highly expressed in human dorsal root ganglion cells and Ia sensory fibers in muscle spindles, often in association with GT1a sugar chains [2]. GQ1b includes two chemical groups composed of two sialic acids, i.e., a disialosyl group, that are bound to galactose, and anti-GQ1b antibodies recognize the sugar chain portion to which sialic acid is bound [2].

Miller Fisher syndrome (MFS) is a clinical syndrome associating ophthalmoplegia, ataxia and areflexia [3].

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Bickerstaff's encephalitis (BE) is a clinical syndrome associating ophthalmoplegia, ataxia and areflexia, i.e., MFS, and a reduced level of consciousness, i.e. drowsiness [4]. BE and MFS have a good outcome and usually occur after an infection, with *Campylobacter jejuni* and *Haemophilus influenzae* being the most frequently identified pathogens [5–7]. As such, both BE and MFS are considered as Guillain-Barré syndrome (GBS) variants [5].

Before anti-GQ1b antibodies were found in both conditions in 1992, MFS and BE were considered separate clinical entities. Today, as MFS and BE share a common serological profile and pathophysiology, both conditions are referred as being part of the “anti-GQ1b antibody syndrome” [6, 7]. Large series of MFS and BE patients with anti-GQ1b antibodies have been described, mainly in Asian countries where both conditions are common [8, 9]. Intriguingly, in these series, 10–30% of patient are GQ1b-seronegative using conventional assays [2, 10]. However, calcium-dependent anti-GQ1b antibodies have been recently found in a majority of these “GQ1b seronegative” patients, thus demonstrating anti-GQ1b antibodies are a major component of MFS and BE pathophysiology [2, 10].

In this retrospective study, we analyzed the clinical, electrophysiological, laboratory and treatment response characteristics of patients with acute anti-GQ1b antibody syndrome originating from 19 French and Swiss University Hospitals. Our aim was to define the clinical and paraclinical features of acute anti-GQ1b-associated neurological syndromes, and to provide clues as to how better manage and treat these patients.

Patients & methods

Patients

We performed an observational retrospective study including patients with: (1) neurological symptoms appearing in less than 1 month; (2) anti-GQ1b antibodies in serum; and (3) diagnosed between 2012 and 2022. Neurologists from 19 University Hospitals in France and Switzerland were contacted, and clinical, paraclinical & treatment data were gathered from medical records by the referring physician. Treatment response was evaluated by the referring neurologist according to each center practice. This study is in conformity with the World Medical Association Declaration of Helsinki 1964 and its later amendments. Given that our study was conducted retrospectively on documented patient records with no specific intervention of any kind, local ethical approval and specific informed consent were not required.

Neurological assessment & diagnostic classification

Data gathering was performed between February and September 2022. In addition to neurological symptoms and signs, data on treatment and outcome were also gathered. Patients were classified according to Shahrizaila's clinical classification [11, 12]: (1) GBS spectrum, including classic GBS and other GBS variant, i.e., paraparetic GBS, pharyngeal cervical brachial weakness (PCBW), facial diplegia and paresthesia; (2) MFS spectrum, including classic MFS and other MFS subgroups, i.e., acute ophthalmoplegia or acute ataxic neuropathy; and (3) BE spectrum, including classic BE & acute ataxic hypersomnolence. If patients had overlapping features between these groups, they were designated as MFS/GBS or GBS/BE.

Biological assessment

Anti-GQ1b antibodies were investigated using different methods in each University Hospital, including commercially available immunodots (GA Generic Assays GmbH, Germany) and ELISA (Enzyme Linked Immunosorbent Assay) (Gangliocombi dosing kit, Bühlman, Switzerland) [13, 14]. Calcium-dependent anti-GQ1b antibodies were not investigated [10]. Serum bacteria and virus serology were investigated using standard methods, including *Campylobacter jejuni*, Epstein-Barr virus (EBV), Cytomegalovirus (CMV) and *Mycoplasma pneumoniae*.

Other paraclinical investigations

Electrodiagnostic studies (EDX) were performed using standard methods. Demyelination was defined using standard criteria, including motor distal latency prolongation, reduction of motor conduction velocity, prolongation of F-wave latency, absent F-waves, motor conduction block, abnormal temporal dispersion, and distal Compound Muscle Action Potential (CMAP) duration prolongation [11]. Brain and spinal cord magnetic resonance imaging (MRI), and cerebrospinal fluid (CSF) analysis were performed using standard methods.

Statistical analysis

We analyzed the clinical and paraclinical characteristics of our patients, including age, sex, clinical subtype (MFS, MFS/GBS, GBS, GBS/BE, and BE), initial clinical severity (as measured with Intensive Care Unit [ICU] admission), anti-GQ1b antibodies serological profile (IgM, IgG, and IgM + IgG), and treatment (PE, IVIg, and CTX) to identify predictors of full recovery at 12 months and predictors of

relapse. Quantitative variables were investigated using Wilcoxon–Mann–Whitney test and Student's t-test, and qualitative variables were investigated using chi-squared and Fisher tests. Two-tailed tests were used with significance level set to 0.05. Statistical analyses were performed using SPSS, Microsoft Excel Software and BiostaTgv.

Results

Neurological characteristics of 166 patients with acute anti-GQ1b antibody syndrome

A hundred and sixty-six (166) patients with acute anti-GQ1b syndrome were identified in the 2012–2022 timeframe in 19 French and Swiss University Hospitals. All patients presented with a monophasic illness with sensory and/or motor symptoms, altered tendon reflexes, reaching nadir within 28 days, in the absence of alternative cause, consistent with GBS. Clinical characteristics of this group of 166 patients are detailed in Table 1. Mean age was 50 years (4–90) and 51% of cases were female. The most frequent clinical signs were abolished tendon reflexes (80% of cases), distal upper and lower limbs sensory symptoms (78%), ophthalmoplegia (68%), sensory ataxia (67%), limb muscle weakness (45%) and bulbar weakness (45%). Facial weakness was observed in 27% of cases, and Babinski sign in 3%. Most patients (69%) had a preceding event before disease onset, including respiratory infection (34% of cases), gastroenteritis (15%), flu-like syndrome (10%), and vaccination (4%). Twenty-eight percent (28%) of patients presented with headache when admitted, and 22% had central nervous system (CNS) involvement, including mainly cerebellar ataxia and more rarely hypersomnolence. One patient (0.6%) died of respiratory failure in ICU.

Clinical classification of patients is detailed in Table 2. Fifty-three patients (32%) had MFS spectrum clinical features, thirty-six patients (22%) had GBS spectrum clinical features, and only one patient (0.6%) had BE spectrum features. Seventy-three patients (44%) presented with mixed clinical features, including sixty-five patients (39%) with MFS/GBS features and eight patients (5%) with GBS/BE features. Three patients were unclassified because of insufficient clinical data.

Biological characteristics of 166 patients with acute anti-GQ1b antibody syndrome

Biological characteristics of our patients are detailed in Table 3. All patients had anti-GQ1B antibodies. In 118 patients in whom IgG was specifically tested, 111 (94%) were positive. In 111 patients in whom IgM was specifically tested, 28 (25%) were positive. Twenty-one patients (13%)

Table 1 Clinical characteristics of 166 patients with acute anti-GQ1b antibody syndrome

Clinical characteristics	N (%)
Female	84 (51%)
Male	82 (49%)
Mean age at disease onset (range)	50 years (4–90)
Preceding event	
Respiratory infection	56 (34%)
No obvious previous infection	52 (31%)
Diarrhea	24 (15%)
Flu-like syndrome	17 (10%)
Angina	8 (5%)
Vaccination	6 (4%)
Abolished tendon reflexes	133 (80%)
Upper limbs	113 (68%)
Lower limbs	129 (78%)
Distal sensory involvement	129 (78%)
Upper limbs	24 (15%)
Lower limbs	8 (5%)
Both	97 (58%)
Proximal sensory involvement	49 (30%)
Upper limbs	8 (5%)
Lower limbs	19 (11%)
Both	22 (13%)
Sensory ataxia	111 (67%)
Proximal weakness	71 (43%)
Upper limbs	10 (6%)
Lower limbs	13 (8%)
Both	48 (29%)
Distal weakness	56 (34%)
Upper limbs	14 (8%)
Lower limbs	5 (3%)
Both	37 (22%)
Babinski sign	5 (3%)
Ophthalmoplegia	113 (68%)
Ptosis	56 (34%)
Pupillary abnormalities	29 (17%)
Mydriasis	24 (14%)
Myosis	2 (1%)
Anisocoria	2 (1%)
Non-reactive pupils	1 (0.6%)
Bulbar impairment	74 (45%)
Dysphagia	56 (34%)
Dysphonia	52 (31%)
Dyspnea	21 (13%)
Facial weakness	45 (27%)
Unilateral	28 (17%)
Bilateral	17 (10%)
Headache	46 (28%)
Central nervous system involvement	37 (22%)
Cerebellar ataxia	28 (17%)
Hyper somnolence	10 (6%)

Table 1 (continued)

Clinical characteristics	N (%)
Confusion	4 (2%)
Spasticity	2 (1%)

Table 2 Clinical classification of 166 patients with acute anti-GQ1b antibody syndrome

Clinical classification	N (%)
MFS spectrum	53 (31.9%)
Typical MFS	35
Incomplete MFS (including acute ophthalmoplegia and acute ataxic neuropathy)	18
MFS/GBS spectrum	65 (39.2%)
MFS/GBS	23
MFS/PCBW	22
MFS/PCBW/FW	8
Incomplete MFS/GBS	5
Incomplete MFS/PCBW/FW	3
MFS/FW	2
Incomplete MFS/PCBW	2
GBS spectrum	36 (21.7%)
Typical GBS	31
PCBW/FW	3
FW	2
GBS/BE spectrum	8 (4.8%)
BE/GBS	7
BE/PCBW	1
BE spectrum	1 (0.6%)
Typical BE	1
Unclassified	3 (1.8%)

BE Bickerstaff encephalitis; FW facial weakness; GBS Guillain Barré syndrome; MFS Miller Fisher syndrome; PCBW pharyngeal-cervical-brachial weakness

had both anti-GQ1b IgG and IgM antibodies. Other anti-ganglioside antibodies were found in 81% of cases, including anti-GT1a antibodies in 56% of patients.

Other paraclinical investigations

Paraclinical characteristics of our patients are detailed in Table 3. CSF analysis was performed on average 5 days after disease onset and showed raised protein content in 24% of cases. In our group of 166 patients, mean CSF protein content was 0.6 g/L. EDX studies were performed on average 9 days after disease onset and were normal in 46% of cases. Demyelinating neuropathy was observed in 28% of cases and axonal neuropathy in 23%. Brain MRI was normal in 106/122 patients (87%). Cranial nerve abnormalities, including hyperintense signals and/or contrast enhancement after

gadolinium injection involving oculomotor, trigeminal, and facial nerve, were observed in 11% of cases. Spinal cord MRI showed myelitis in two patients, including 1 cervical and 1 thoracic. Serological proof of recent infection was found in 17% of cases, including *Campylobacter jejuni* (7% of cases), *Mycoplasma pneumoniae* (4%), Epstein Barr virus (2%) and Cytomegalovirus (2%).

Treatment and outcome of 166 patients with acute anti-GQ1b antibody syndrome

Fifty-three patients (32%) were admitted in ICU, and almost half of them (26; 16%) required mechanical ventilation. One patient died in ICU from respiratory failure. Most patients (148 patients; 89%) were treated, including with intravenous immunoglobulins (IVIg) (138 patients; 83%), corticosteroids (CTX) (17 patients; 10%) and plasma exchange (PE) (9 patients; 5%). Amongst the 138 patients treated with IVIg, 124 were treated with one dose of 2 g/kg administered over 5 days, 13 with 2 doses of 2 g/kg, and 1 with 3 doses of 2 g/kg. The 17 patients treated with CTX were given IV methylprednisolone 1 g/day over 1 to 3 days in all cases. Five PE sessions were done in the 9 patients treated with PE. No maintenance therapy was given in the cases described here. After a mean 12 months follow-up, 69% of patients presented with complete neurological recovery and 29% with incomplete neurological recovery. Thirteen patients (8%) had presented with a previous acute neurological episode before being included in our series, and 12 patients (7%) presented with an acute anti-GQ1b antibody syndrome relapse after the episode described here. Overall, 25/166 patients (15%) from our series had a relapsing history.

Predictors of full recovery at 12 months and predictors of relapse

We identified several predictors of complete recovery at 12 months, including age < 70 years (74% of patients with age < 70 years recovered completely vs. 45% of patients with age > 70 years, $p < 0.001$), no initial ICU admission (50% of patients who had not been admitted in ICU recovered completely vs 27% of patients who had been admitted in ICU, $p < 0.008$) and the presence of anti-GQ1b IgG antibodies, whether associated or not with IgM antibodies (73% of patients with anti-GQ1b IgG antibodies recovered completely vs 28% of patients with only anti-GQ1b IgM antibodies, $p = 0.02$). Patients with MFS tended to more frequently fully recover than patients with MFS/GBS, GBS and GBS/BE, but the comparison between the MFS subgroup and other subgroups did not reach statistical significance (80% of patients with MFS recovered completely vs 64% of patients with MFS/GBS, GBS and GBS/BE, $p = 0.08$). Importantly, treatment was not a predictor of full recovery.

Table 3 Paraclinical characteristics of 166 patients with acute anti-GQ1b antibody syndrome

Paraclinical characteristics	N (%)
Serologic evidence of recent infection	
No evidence of infection	138/166 (83%)
<i>Campylobacter jejuni</i>	12/166 (7%)
<i>Mycoplasma pneumoniae</i>	6/166 (4%)
Epstein-Barr virus	4/166 (2%)
Cytomegalovirus	3/166 (2%)
Cerebrospinal fluid (CSF)	
Raised protein content	38/159 (24%)
Mean protein content (g/l)	0.6
Mean CSF white cells (/mm ³)	10
Mean delay between disease onset and CSF analysis (days)	5
Nerve conduction studies (NCS)	
Normal	70/152 (46%)
Demyelinating	43/152 (28%)
Axonal	35/152 (23%)
Insufficient data	4/152 (3%)
Mean delay between disease onset and NCS (days)	9
Brain magnetic resonance imaging (MRI)	
Normal	106/122 (87%)
Abnormal	16/122 (13%)
Cranial nerve abnormalities	12/122 (10%)
White matter abnormalities	3/122 (2%)
Cranial nerve & white matter abnormalities	1/122 (1%)
Anti-ganglioside antibodies	
IgG anti-GQ1b	111/118 (94%)
IgM anti-GQ1b	28/111 (25%)
IgG & IgM anti-GQ1b	21/166 (13%)
Other anti-ganglioside antibodies	134/165 (81%)
Anti-GT1a antibodies	92/165 (56%)

We did not identify predictors of relapse: age, sex, initial clinical severity, clinical subtype, outcome at 12 months and anti-GQ1b antibodies serologic profile did not significantly differ between the group of patients with relapses and the group of patients with no relapses.

Discussion

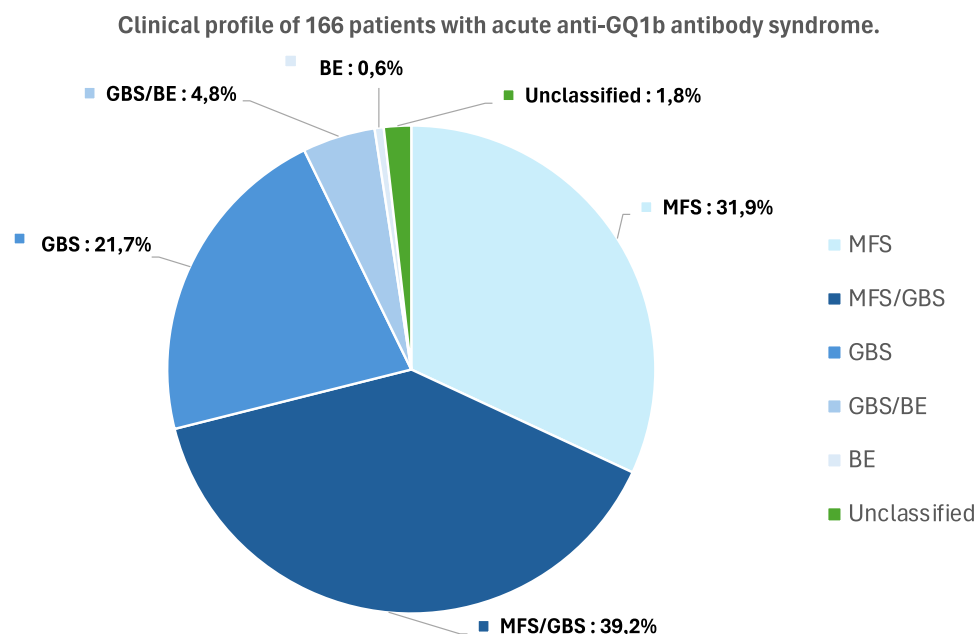
In this retrospective study we analyzed the clinical, EDX, biological, treatment response and outcome characteristics of 166 patients with acute anti-GQ1b antibody syndrome. To our knowledge, this is the largest cohort from Western Europe described until now.

The clinical spectrum of acute anti-GQ1b antibody syndrome is not well known. Indeed, in contrast with our study, a majority of previous studies analyzed patients on the basis of clinical presentation, i.e., MFS and/or BE, and not on the basis of the presence of anti-GQ1b antibodies in serum [9]. In addition, most studies describing MFS originates from

Asia, and there is no major study analyzing acute anti-GQ1b syndrome in a European population.

Our study shows that MFS clinical presentation is observed in only 32% of patients with acute anti-GQ1b syndrome, while 44% of patients present with mixed MFS/GBS & GBS/BE clinical features (Fig. 1). This observation suggests the phenotypical spectrum of acute anti-GQ1b is large, ranging from typical MFS to typical BE, with most patients presenting with mixed phenotypes including typical GBS presentation. Only 1 patient (0.6%) in our series presented with typical BE, highlighting the fact that BE is an extremely rare condition. Our study also shows that acute anti-GQ1b syndrome tends to recur, with almost 1 out of 7 patients (15%) presenting with relapses. Relapsing acute anti-GQ1b syndrome has already been described, and no predisposing factor has been identified until now [15–17]. Our study confirms the latter, as we failed to identify predictors of relapse, including age, sex, initial clinical severity, clinical subtype, outcome at 12 months, and anti-GQ1b antibodies serologic profile. Our study confirms acute anti-GQ1b antibody

Fig. 1 Clinical spectrum of 166 patients with acute anti-GQ1b antibody syndrome. Fifty-three patients (31.9%) presented with complete (21.1%) and incomplete (10.8%) Miller Fisher syndrome (MFS), thirty-six (21.7%) with Guillain-Barré syndrome (GBS), one (0.6%) with Bickerstaff encephalitis (BE), and seventy-three (44%) with mixed MFS/GBS & GBS/BE clinical features. *BE* Bickerstaff encephalitis; *GBS* Guillain-Barré syndrome; *MFS* Miller Fisher syndrome



syndrome has low mortality (0.6%), and demonstrates CNS involvement is significant, including spinal cord MRI abnormalities in 2 cases and brain MRI abnormalities in 13% of patients.

In our study, 56% of patients presented with anti-GT1a antibodies in addition to anti-GQ1b antibodies. This result is in accordance with previous results demonstrating that almost all patients with anti-GQ1b IgG antibodies also present with an anti-GT1a IgG activity [18, 19]. Absorption studies suggest that the same antibody reacts with GQ1b and GT1a [18, 19].

In our study, 46% of patients had normal EDX investigations, 28% had demyelinating neuropathy, and 23% had axonal neuropathy. It is interesting to note that EDX investigations in patients with MFS and BE usually show a pattern of axonal sensory neuropathy with no demyelination [20, 21]. This apparent discrepancy is likely explained by the fact that we studied a group of patients with acute anti-GQ1b antibody syndrome and not a group of patients with MFS. Most patients in our study (66% of cases) presented with GBS-like clinical phenotypes, and EDX findings in our study likely reflect this clinical heterogeneity. This discrepancy may also be in part explained by the fact that EDX investigations were performed early in our study, i.e., after a mean delay of 9 days after symptoms onset, before significant axonal loss and/or demyelination is likely to appear.

MFS and BE are usually self-limiting disorders with good spontaneous prognosis [7]. In this context, whether patients with MFS and BE require immunomodulating treatment remains unclear. A 2007 Cochrane review found no randomized data for the treatment of MFS and BE [22]. However, the authors proposed to extrapolate the evidence

for treating GBS to both MFS and BE [22]. Interestingly, a recent retrospective study found treated and untreated MFS patients had similar outcome [23]. In this context, one may hypothesize treatment may be useful in patients with a severe GBS-like clinical phenotype [24], and perhaps less relevant in patients with a typical MFS phenotype [25]. In our series, 83% of patients were treated with IVIg, and outcome was excellent in most cases. However, when considering predictors of full recovery at 12 months in our series, we observed patients with age < 70 years, MFS clinical presentation, no ICU admission and anti-GQ1b IgG antibodies more frequently presented with complete recovery at 1 year in comparison with other patients, i.e., patients with age > 70 years, GBS clinical presentation, initial ICU admission and no anti-GQ1b IgG antibodies. Interestingly, treatment was not a predictor of full recovery. In this context, whether IVIg treatment was relevant in our series remains unclear. Additional studies are needed to determine whether immunomodulating therapies are useful in all patients or only a subset of patients with acute anti-GQ1b antibody syndrome.

Pathological data suggest MFS is a nodo-paranodopathy likely involving a complement-mediated disruption of the node of Ranvier [26]. When considering CNS involvement, it has been shown that sera from BE patients destroyed the blood–brain-barrier (BBB), which was not the case with sera from patients with MFS. BE serum-related BBB disruption may be the result of autocrine secretion of metalloproteinase MMP-9 from BBB-composing endothelial cells [27].

Our series has limits. First, the retrospective design led to missing data, including detailed clinical and EDX follow-up. Second, the multicentric design led to absent

standardized data collection and clinical evaluation. Third, we did not have data detailing how many patients with no polyneuropathy presented with anti-GQ1b antibodies in our 19 recruiting centers. Nonetheless, it is known that the incidence of anti-ganglioside antibodies in healthy individuals ranges from 1 to 9% [28–30]. Fourth, the fact that anti-GQ1b antibodies were investigated using different methods in each University Hospital did not allow us to investigate correlations between antibodies titers and other variables such as outcome, relapse, and clinical presentation.

Conclusion

To summarize, our study demonstrates acute anti-GQ1b antibody syndrome presents with a wide GBS-like clinical spectrum mostly involving PNS and rarely CNS. Mortality is low, outcome is good in 2/3 of cases, and relapses are frequent. Most patients in our study were treated with IVIg, but whether this treatment is useful in all cases of acute anti-GQ1b antibody syndrome remains to be evaluated.

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Data availability All data and related documentation from this study are available on request to qualified researchers with no time limit, and subject to a standard data-sharing agreement.

Declarations

Conflict of interest MC, DA, SA, FB, JPC, GC, CC, JBC, PC, AC, ED, GF, SF, TG, TK, CL, TM, MMi, MMo, GN, JBN, YP, AP, GS, FT, CT, MT, ST, LV & AEL disclose all competing interests. Dr M. Coly, Pr D. Adams, Pr S. Attarian, Dr F. Bouhour, Pr JP. Camdessanche, Dr G. Carey, Dr C. Cauquil, Dr P. Chretien, Dr E. Delmont, Dr G. Fargeot, Dr S. Frachet, Dr T. Gendre, Dr T. Kuntzer, Dr C. Labeyrie, Dr T. Maisonnobe, Dr M. Michaud, Dr M. Moulin, Pr G. Nicolas, Dr JB. Noury, Pr Y. Pereon, Dr A. Puma, Dr G. Sole, Dr F. Taithe, Dr C. Tard, Dr M. Theaudin, Pr S. Timsit, Dr L. Venditti: declares no disclosures, Dr JB. Chanson declares consultancy fees from Alnylam. Pr A. Creange received grants from Octapharma. Pr A. Echaniz-Laguna received consulting fees from LFB, Alnylam, and Pfizer.

Ethical standards For this type of study formal consent is not required.

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